Recap: Divergent sequences - Types.

Convergent => Bounded. ↓ Contrapositive

Unbounded => Divergent.

Take a sequence (a_n) and another sequence (b_n) we can define the following:

1) $\lambda(a_n) = (\lambda a_n)$ where $\lambda \in \mathbb{R}$.

Multiplication on $\mathbb{R}$

2) $(a_n) + (b_n) = (a_n + b_n)$

3) $(a_n)(b_n) = (a_n b_n)$

4) $\dfrac{(a_n)}{(b_n)} = \left(\dfrac{a_n}{b_n}\right)$.

Theorem 6.1: Let (a_n) and (b_n) be convergent sequences with $\lim(a_n) = a$ and $\lim(b_n) = b$.

Then it follows that

i) $\lim(ca_n) = ca$ for all $c \in \mathbb{R}$

ii) $\lim(a_n + b_n) = a + b$

iii) $\lim(a_n b_n) = ab$

iv) $\lim\left(\frac{a_n}{b_n}\right) = \frac{a}{b}$ providing $b \neq 0$ and $b_n \neq 0$ $\forall n \in \mathbb{N}$.

Proof: i) Since $\lim(a_n) = a$, it follows that for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t

$$|a_n - a| < \varepsilon/|c|$$

for every $n > N$. Hence, for all $n > N$

$$|ca_n - ca| = |c||a_n - a|$$

$$< |c| \varepsilon/|c|$$

$$= \varepsilon.$$

Exercise for when $c = 0$. ∎

ii) Scratch Work: I'm able to control $|a_n - a| < \varepsilon$ ($n > N_1$) and $|b_n - b| < \varepsilon$ ($n > N_2$). Want $|(a_n + b_n) - (a + b)| < \varepsilon$ ($n > N$).

Realise that $|(a_n+b_n)-(a+b)| \leq |a_n-a|+|b_n-b|$

$< \varepsilon/2 + \varepsilon/2$

$= \varepsilon.$

Proof: Since $\lim(a_n) = a$ it follows that given $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ s.t

$$|a_n - a| < \varepsilon/2$$

for all $n > N_1$. Similarly, since $\lim(b_n) = b$, there exists $N_2 \in \mathbb{N}$ s.t

$$|b_n - b| < \varepsilon/2$$

for all $n > N_2$. Define $N = \max(N_1, N_2)$. Then for all $n > N$ we have that

$$|(a_n+b_n) - (a+b)| \leq |a_n-a| + |b_n-b|$$

$$< \varepsilon/2 + \varepsilon/2$$

$$= \varepsilon. \quad \square$$

(iii) Scratch Work: Control $|a_n-a| < \varepsilon$ and $|b_n-b| < \varepsilon$.

Want $|a_nb_n - ab| = |a_nb_n - a_nb + a_nb - ab|$

$$= |a_n(b_n - b) + b(a_n - a)|$$

$$\leq \underbrace{|a_n|}_{< M}\underbrace{|b_n - b|}_{< \frac{\varepsilon}{2M}} + |b|\underbrace{|a_n - a|}_{< \frac{\varepsilon}{2|b|+1}}$$

Since Convergent $\Rightarrow$ Bounded

Proof: Since $\lim(a_n) = a$, it follows that for all $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ s.t

$$|a_n - a| < \frac{\varepsilon}{2|b| + 1}$$

for all $n > N_1$. Since (a_n) is convergent, it is also bounded. Hence, there exists $M > 0$ s.t $|a_n| < M$ for all $n \in \mathbb{N}$. Since $\lim(b_n) = b$ it follows that there exists $N_2 \in \mathbb{N}$ s.t

$$|b_n - b| < \frac{\varepsilon}{2M}$$

for all $n > N_2$. Define $N = \max(N_1, N_2)$. Then for all $n > N$ we have that

$$|a_n b_n - ab| = |a_n b_n - a_n b + a_n b - ab|$$

$$\leq |a_n||b_n - b| + |b||a_n - a|$$

$$< M \cdot \frac{\varepsilon}{2M} + |b| \frac{\varepsilon}{2|b|+1}$$

$$= \frac{\varepsilon}{2} + |b| \frac{\varepsilon}{2|b|+1}$$

$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

$$= \varepsilon. \quad \text{囧}$$

(iv) Do the scratch work yourself and compare with the proof in the notes.

Ex 6.2 : (Side: Since $1/n \to 0$ we now have $1/n^p \to 0$ $\forall n \geq 1$, $n \in \mathbb{N}$)

i) Say $\lim\left(\frac{1}{n} + \frac{4}{n^2}\right) = \lim\left(\frac{1}{n}\right) + \lim\left(\frac{4}{n^2}\right)$

$$= \lim\left(\frac{1}{n}\right) + 4\lim\left(\frac{1}{n}\right) \cdot \lim\left(\frac{1}{n}\right)$$

$$= 0 + 4(0)(0)$$

$$= 0.$$

ii) Take (a_n) to be sequence

$$a_n = \frac{3n^2 - 6n + 2}{5n^2 - 2n + 3}.$$

Divide top and bottom by the largest power of n to give

$$a_n = \frac{3 - 6/n + 2/n^2}{5 - 2/n + 3/n^2}.$$

Then $\lim(a_n) = 3/5$.

Theorem 6.3: Let (a_n) and (b_n) be convergent sequences with $\lim(a_n) = a$ and $\lim(b_n) = b$. Then,

i) If $a_n \geq 0$ for all $n \in \mathbb{N}$, then $a \geq 0$.

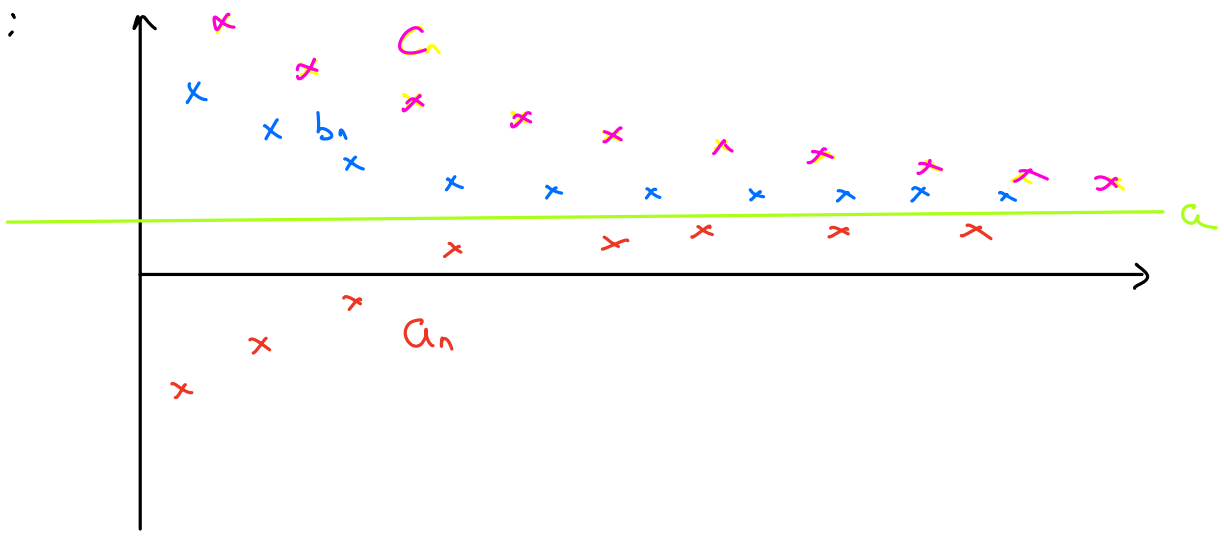
ii) If $a_n \leq b_n$ for all $n \in \mathbb{N}$, then $a \leq b$.

iii) If there exists a $c \in \mathbb{R}$ satisfying $c \leq b_n$ for all $n \in \mathbb{N}$, then $c \leq b$. Similarly, if $a_n \leq c$

for all $n \in \mathbb{N}$, then $a \leq c$.

Proof: (i) See notes. (ii)(iii) - Exercises.

Picture:



Theorem 6.4: Squeeze Theorem

If $a_n \leq b_n \leq c_n$ for all $n > m$ and $\lim(a_n) = \lim(c_n) = a$. Then $\lim(b_n) = a$.

Proof: For every $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ s.t

$$|a_n - a| < \varepsilon \quad (\Leftrightarrow a - \varepsilon < a_n < \varepsilon + a)$$

for all $n > N_1$. Similarly, there exists $N_2 \in \mathbb{N}$ s.t

$$|c_n - a| < \varepsilon \quad (\Leftrightarrow a - \varepsilon < c_n < a + \varepsilon)$$

for all $n > N_2$. Let $N = \max(m, N_1, N_2)$. Then

for every $n > N$, we have that

$$a - \varepsilon < a_n \leq b_n \leq c_n < a + \varepsilon$$

$$\Rightarrow \quad a - \varepsilon < b_n < a + \varepsilon$$

$$\Rightarrow \quad |b_n - a| < \varepsilon . \quad \square$$

Ex: (i) Let (a_n) be the sequence given by

$$a_n = \frac{\sin(n)}{n} .$$

Since $-1 \leq \sin(n) \leq 1$ for every $n \in \mathbb{N}$ it follows that

$$-\frac{1}{n} \leq \frac{\sin(n)}{n} \leq \frac{1}{n}$$

for every $n \in \mathbb{N}$. Hence by the squeeze theorem,

$\lim\left(\frac{\sin(n)}{n}\right) = 0$. $\square$

(ii) Let (a_n) be a sequence given by $a_n = \frac{x^n}{n!}$

for fixed positive $x \in \mathbb{R}^+$. Since $n! > 0$ for all $n \in \mathbb{N}$ and $x > 0$ it follows that

$$0 \leq x^n/n! \cdot \left(\frac{x^n}{n!} = \overbrace{\frac{x}{1} \cdot \frac{x}{2} \cdots \frac{x}{n-1} \cdot \frac{x}{n}}^{n \text{ times}} \right)$$

Let $m = \min\left(\{ n \in \mathbb{N} : n \geq x \}\right)$. For $n > m$, we have that

$$\frac{x^n}{n!} = \frac{x}{1} \cdot \frac{x}{2} \cdots \frac{x}{m-1} \cdot \frac{x}{m} \cdots \frac{x}{n-1} \cdot \frac{x}{n}$$

$$\leq \frac{x}{1} \cdot \frac{x}{2} \cdots \frac{x}{m-1} \cdot \frac{x}{n}.$$

Define $c = \frac{x}{1} \cdot \frac{x}{2} \cdots \frac{x}{m-1}$. Hence, for all $n > m$ we have that

$$0 \leq \frac{x^n}{n!} \leq \frac{cx}{n}.$$

Since $\lim\left(\frac{cx}{n}\right) = 0$, by the Squeeze Theorem it follows that $\lim\left(\frac{x^n}{n!}\right) = 0$. ∎

So far, we are only able to prove a convergent sequence is convergent, if we can 'guess' the value it converges to, or sandwich it between two other known convergent sequences.

Q: Do there exist sequences which are convergent and we are unable to 'guess' the value they converge to?

A: Consider

$$a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}, \text{ with } a_1 = 1.$$

This is a monotone sequence and we suspect that if a sequence is monotone and bounded it should be convergent. → Monotone Convergence Theorem.